

1 Intro to R and R markdown

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Question 1

What is allowed or not allowed with variable names? **Try assigning values to the following variables. Which work and which do not? What is causing the errors?**

For variable names in R:

- allowed
 - starts with a letter or period not followed by a number.
 - can include letters, periods, numbers, and underscores.
- not allowed
 - start with an underscore
 - start with a number

```
min_height <- 1 #works
max.height <- 1 #works
_age <- 1 #Error: unexpected symbol in "_age"
.mass <- 1 #works, creates hidden variable
MaxLength <- 1 #works
min-length <- 1 #Error in min - length <- 1 : could not find function "-<-"
2widths <- 1 #Error: unexpected symbol in "2widths"
celsius2kelvin <- 1 #works
```

In the assignment operations above `_age <- 1` throws an error since it begins with an underscore, `2widths <- 1` since it begins with a number, and `min-length <- 1` since it contains `-` so it is interpreted as a subtraction operator.

Question 2

What aspects of the “Good” code help you understand the script?

Some aspects of code that help understand a script include:

- Inserting comments (with `#`) to explain a piece of code.
- Adding detailed headers that explain the objective of a script and how to use it.
- Being explicit about code dependencies by loading all packages at top of script and limiting hard-coding of input and output files for a script.
- Choosing descriptive names for variables.
- Separate distinct portions of code into sections.

Question 3

What two properties are common to all of the vectors above?

Two properties common to all vectors are:

1. They have a length (greater than or equal to 0).
2. They contain elements of the same data type.

Question 4

Consider the following matrix:

```
FOURS <- matrix(  
  c(4, 4, 4, 4),  
  nrow = 2,  
  ncol = 2)
```

Given that `typeof(FOURS[1])` returns “double”, what would you expect `typeof(FOURS)` to return? How do you know this is the case even without running this code?

We would expect `typeof(FOURS)` to return “double” since the elements in a matrix must be of the same data type.

Question 5

```
2+2
```

```
## [1] 4
```

Personal reflection

Insert ~3-4 sentences of your personal take on the in-class or group discussion.

Whiteboard question

Instructions: Place your photo in the same folder as this Rmd file. Remove the “eval=F” in the curly braces of the code chunk, and replace the file name with your photo.